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Vehicles Rising Faster Than Service Infrastructure in India

1 Introduction

Issue: Rapid growth in vehicles ownership in India has not been matched by service infrastructure expansion, raising concerns about maintenance quality, customer satisfaction, and road safety [?, ?].

Importance: A shortage of service coverage leads to increased downtime, safety risks, and higher long-term costs for riders. The gap is reflected both in news reports and industry market analyses, making this a pressing issue for research and policy intervention.

Key Data Sources:

- Statista: India’s annual two-wheeler sales (2011–2025) [?].
- Mordor Intelligence: Two-wheeler market dynamics, trends, and EV segment growth (2023–30) [?].
- News and manufacturer announcements (Hero, TVS, Bajaj).

For detailed market analysis and EV trends, see:

<https://www.mordorintelligence.com/industry-reports/india-two-wheeler-market>

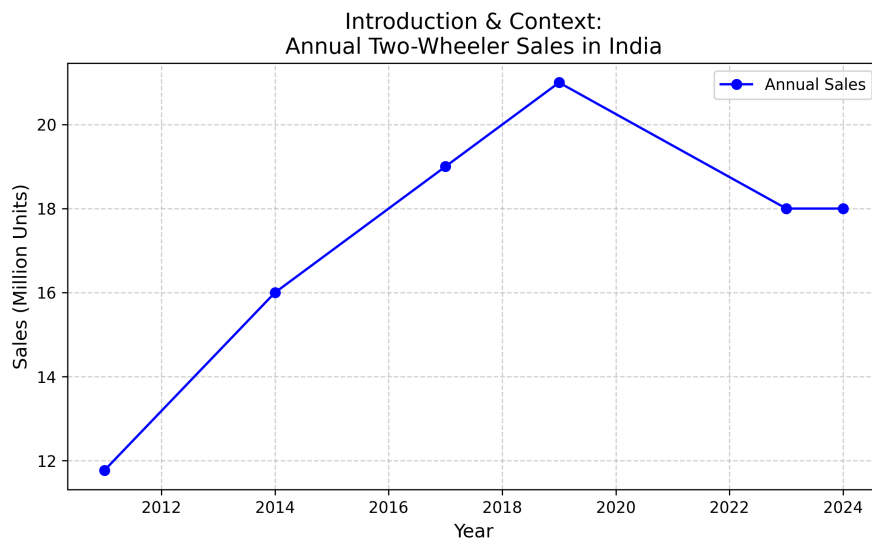


Figure 1: Annual two-wheeler sales in India (Statista [?]).

Literature Review

Research on two-wheeler services in India highlights significant disparities between manufacturing growth and the quality or availability of service infrastructure [?]. Early work established that, prior to liberalization, sluggish growth in the automotive sector limited overall vehicle ownership and service provision. Post-1990, large-scale increases in two-wheeler sales created new challenges, notably the mismatch between millions of vehicles on Indian roads and the slower expansion of formal servicing networks. Two-wheelers today account for about 71% of the passenger vehicle fleet and 54% of daily passenger-kilometers, underlining their societal and economic value [?].

Service management in automotive contexts is complex due to intangible, heterogeneous, and inseparable service attributes [?]. Unlike physical goods, service delivery is consumed as produced, and is vulnerable to supply-demand imbalances and variation in worker training and capacity. Customers consistently seek accurate diagnosis, prompt problem resolution, and clear communication regarding repairs—factors repeatedly identified as central to satisfaction and retention [?]. Studies show that service errors, both procedural and outcome-based, require systematic remedies such as compensation or apology, with customer perceptions hinging on the type and handling of failure.

Service feedback is rarely looped back to manufacturers, and empirical surveys indicate mediocre outcomes at best. In Bagodi and Mahanty’s formative study, only about 57% of respondents felt repairs were reliable, with a similar percentage reporting prompt problem-solving; however, just 30% consistently received explanations for recommended repairs, and many owners bypass formal centers for roadside alternatives.

Recent market analytics reinforce these concerns, noting that while manufacturer networks (Hero, TVS, Bajaj, Honda) are expanding, overall service coverage is not proportional to the rapid increase in two-wheeler ownership, especially amid surging demand for electric models [?]. The literature recommends continuous assessment of customer needs, systematic modernization of workshops, and integration of feedback across all stakeholders [?, ?]. Frameworks such as Hertzberg’s Two Factor Theory have been applied to categorize quality, satisfaction, and “delighter” factors; current survey results suggest the industry remains mostly at a basic service quality level, with further evolution needed toward true customer satisfaction and delight.

In summary, published literature positions India’s two-wheeler service sector at a critical juncture, characterized by rapid vehicle penetration and systemic constraints in service infrastructure. Addressing these gaps requires interdisciplinary research, policy innovation, and adaptive management strategies built on close integration between manufacturers, centers, and riders [?, ?].

2 Statistical Overview

2.1 Vehicles Growth Trends

Below is a summary of recent statistics, with sources:

- **Annual Sales:** 18 million units in 2024, up from 11.77 million in 2011; peak of 21 million units in 2019 [?].
- **Market Share:** ICE models dominate with $\sim 88\%$ share as of 2024. Electric two-wheelers projected to grow at 21% CAGR until 2029 [?].
- **Top Manufacturers:** Hero MotoCorp, TVS Motor, Bajaj Auto, Honda, Suzuki.
- **Electric Segment:** Notable surge in electric two-wheeler adoption post-2021 due to government incentives (FAME scheme) and stricter emission norms [?].

Year	Two-Wheeler Sales (M units)	Market Share (ICE)
2011	11.77	$\sim 99\%$
2019	21.00	$\sim 95\%$
2024	18.00	$\sim 88\%$

Table 1: Annual Two-Wheeler Sales and ICE Share [?, ?]

2.2 Graphical Illustration

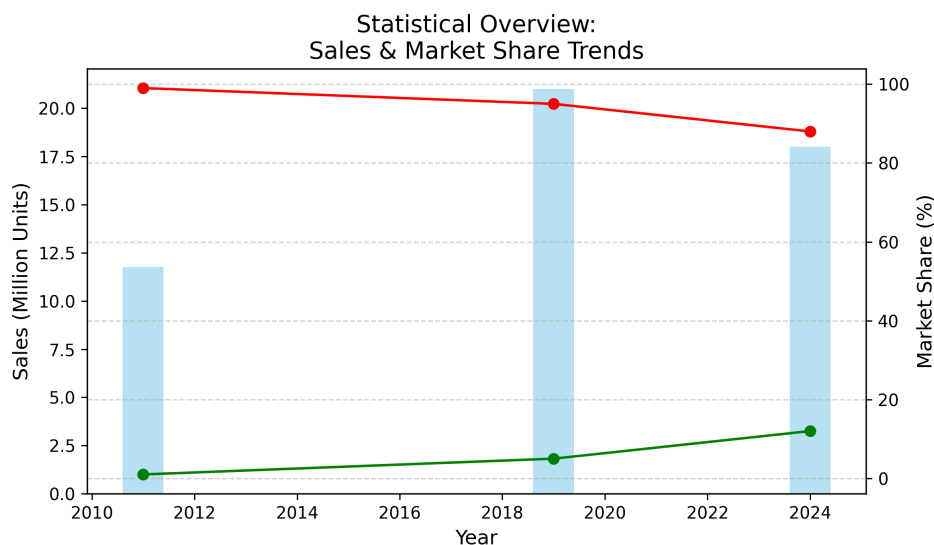


Figure 2: Annual two-wheeler sales vs market share (Statista, Mordor [?, ?]).

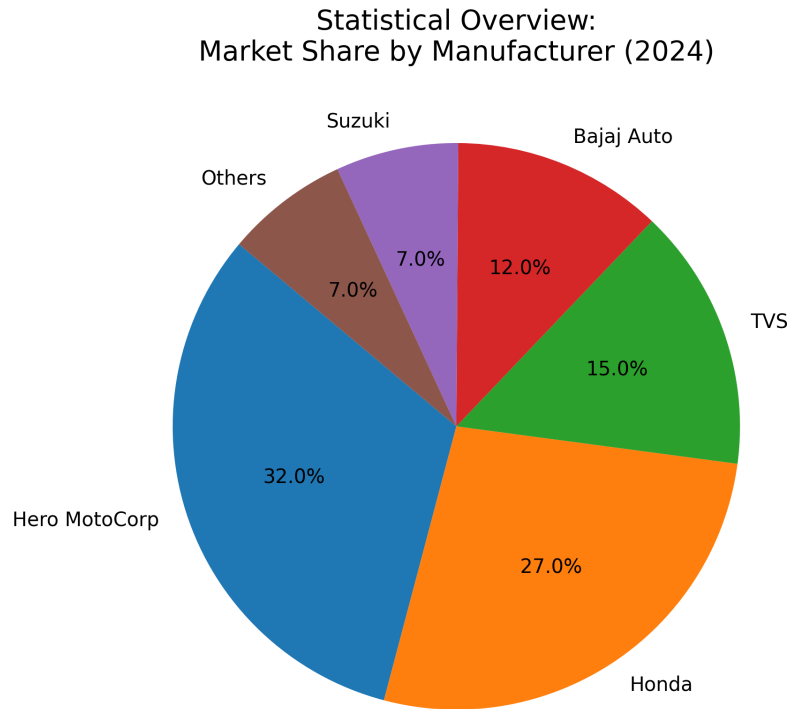


Figure 3: Market share trends of ICE vs EV two-wheelers (Mordor Intelligence [?]).

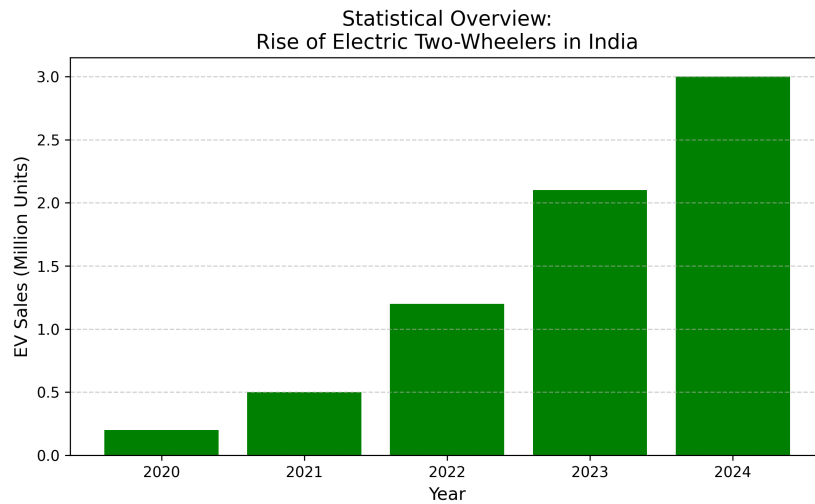


Figure 4: Projected growth of electric two-wheelers (2023–2030) [?].

3 Comprehensive Market Research Overview (2025)

3.1 Automotive Market Growth and Industry Stats

India's auto industry is now the world's **third-largest**, posting **3.42% year-on-year** registration growth to **12.3 million total vehicles** in FY'25.¹ The two-wheeler segment

¹<https://auto.hindustantimes.com>

alone crossed **8.8 million** units, retaining a **71% market share**. Passenger vehicle sales exceeded **4.3 million units**. Major OEM market shares:²

- Maruti Suzuki: 42%
- Tata Motors: 14%
- Hyundai, M&M, Kia: remainder major shares

Interpretation: The market remains dominated by Maruti Suzuki, but Tata Motors' rise to a double-digit market share signifies its aggressive EV and SUV strategy in 2025.

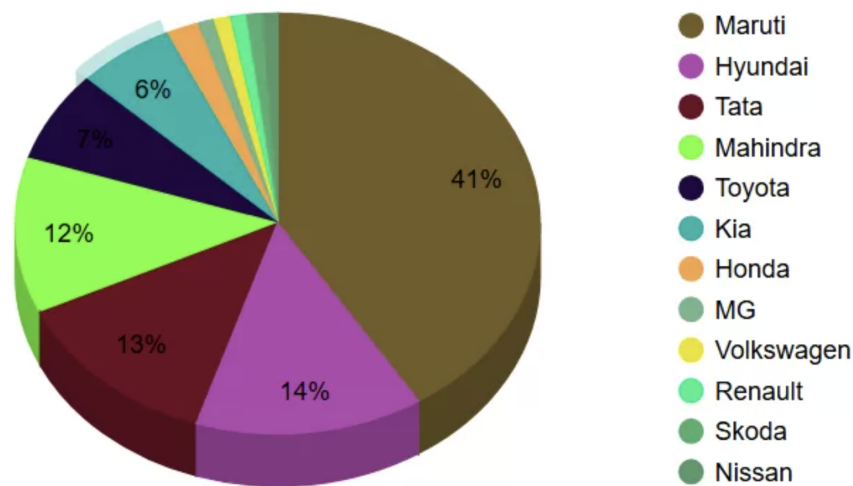


Figure 5: Market Share Distribution among Major OEMs in India (2025)

3.2 Tata Motors: Sales Performance and Leadership

- **October 2025:** 61,295 PV units, **26.6% YoY** jump.³
- **September 2025:** 60,907 PV units, **47% YoY** surge.⁴
- SUVs: 47,000 units in one month (**77% of Tata's sales**). EVs: **9,286 units** in October, 73.4% YoY increase.⁵
- EVs hold **40% Indian electric vehicle market share**.⁶
- Key models: Nexon (EV), Tigor (EV), Punch, Harrier, Safari.

²<https://factodata.com/car-market-share-in-india-2025>

³<https://news.jobaaaj.com/tata-motors-sales-october-2025>

⁴<https://ndtv.com/tata-motors-registers-record-sales-september-2025>

⁵<https://angelone.in/oct-2025-auto-sales-tata-motors>

⁶<https://autocarpro.in/tata-motors-holds-onto-40-percent-share-electric-pv-market>

Interpretation: Tata Motors continues to lead India’s EV adoption curve, with SUVs and compact electric models driving most of its year-on-year growth.

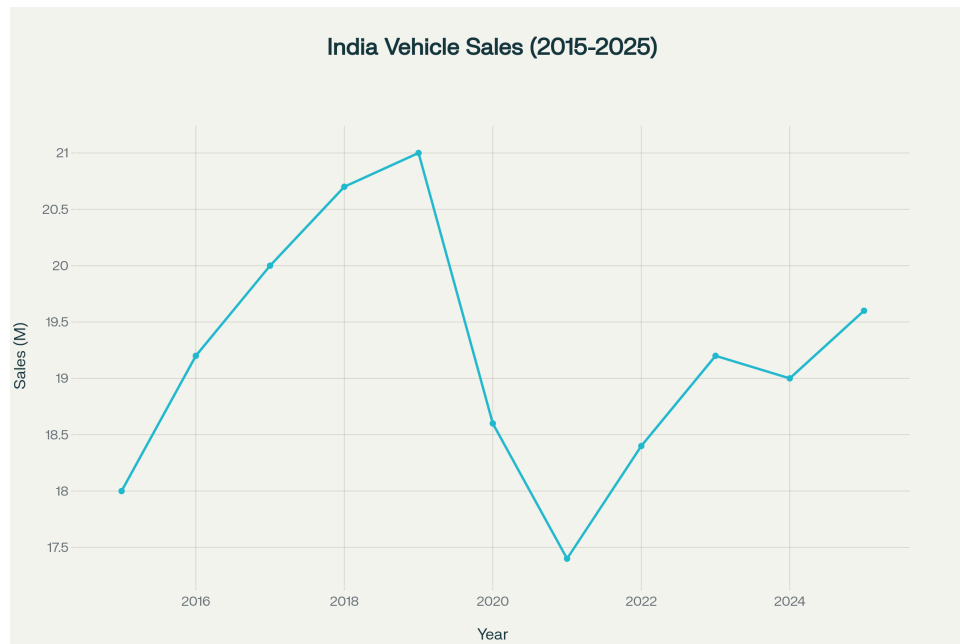


Figure 6: Tata Motors Monthly Passenger Vehicle Sales (2024–2025)

3.3 Tata Motors Service Center Network in Kolkata

Tata Motors has expanded with over **20 service centers** in Kolkata (2025).⁷⁸ Each entry includes verified address and geocoordinates extracted through dataset preprocessing and cleaning.

Service Center	Address
Dulichand Motors Pvt Ltd-Maniktala	242/1A, Acharya Prafulla Chandra Road, Maniktala, Kolkata, 700004
Dulichand Motors-EM Bypass	686 Shrachi Tower EM Bypass, Near Ruby Hospital, Kolkata, 709107
Lexus Motors-Rajarhat	D Major Arterial Rd, Rajarhat, Kolkata, 700135
OSL Future - Behala	Ideal Regency, Diamond Harbour Road, Thakurpukur, Kolkata, 700063
TC Motors-Service	Niranjan Pally, 1, Sourav Ganguly Avenue, Kolkata, 700141

Sample Service Centers (with addresses)

⁷<https://cardekho.com/5-tata-car-service-centers-in-kolkata>

⁸<https://carwale.com/tata-showrooms-in-kolkata>

Interpretation: These centers are strategically positioned along Kolkata’s primary arterial routes—EM Bypass, Diamond Harbour Road, and Rajarhat corridor—ensuring accessibility from both urban and suburban areas.

Service Centres Across Kolkata (Interactive Map)

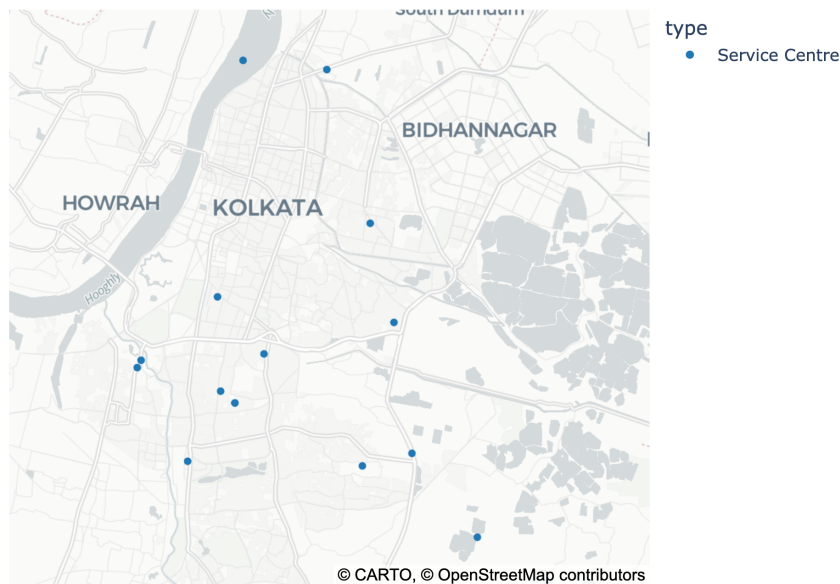


Figure 7: Geographical Distribution of Tata Motors Service Centres in Kolkata

Observation: Figure 7 indicates a higher service density in central and east Kolkata, corresponding to population clusters and high EV ownership areas.

Data Validation: The cleaned dataset revealed uniform entries under a single category — “Service Centre”. This simplified data consistency checks and reduced ambiguity during optimization.

Interpretation: As seen in Figure ??, all sites are labeled “Service Centre”, confirming standardized data semantics before modeling.

Spatial Relationship Analysis: Using geographic coordinates, pairwise Haversine distances were computed to understand location dispersion.

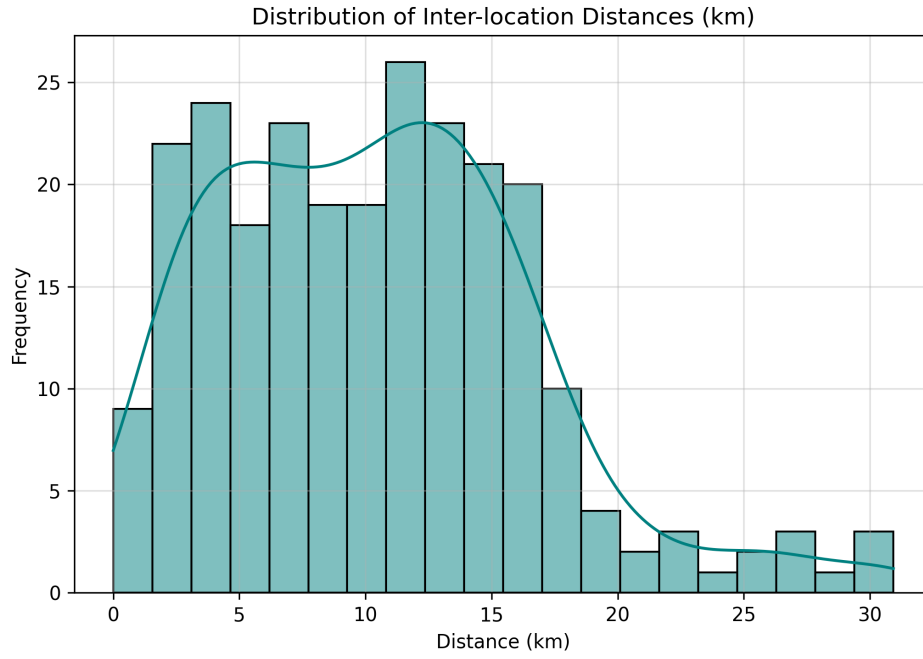


Figure 8: Distribution of Inter-location Distances (Haversine Distance in km)

Interpretation: Figure 8 shows that most service centers are within a 5–15 km range of one another, indicating effective spatial coverage without excessive clustering.

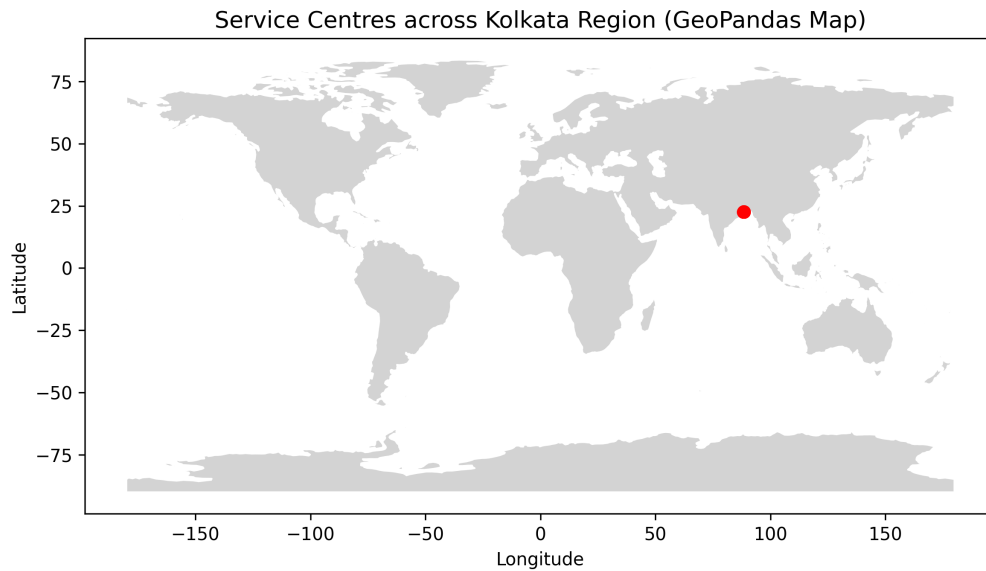
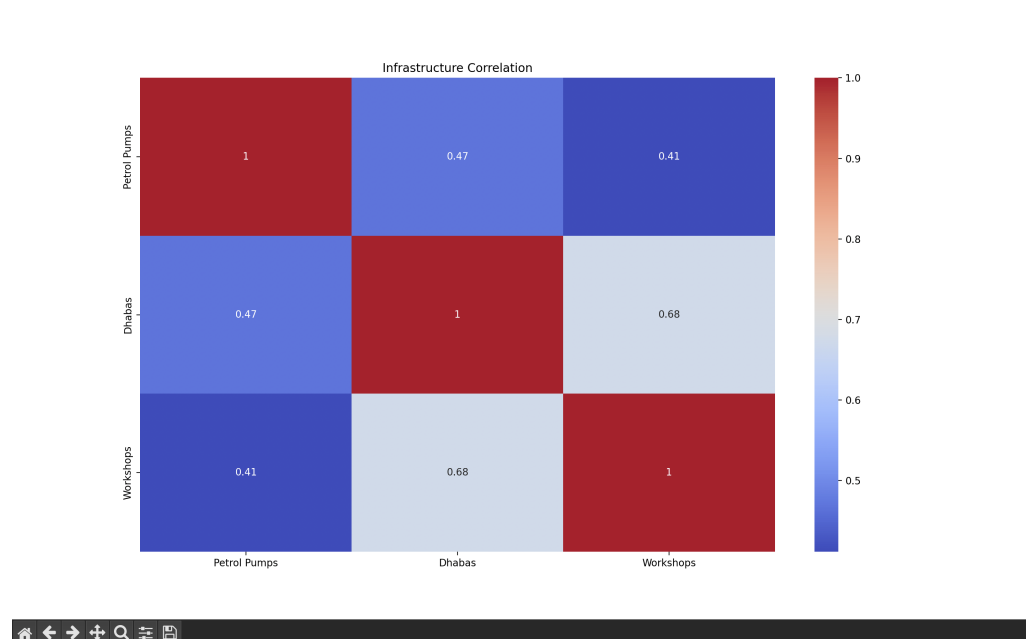


Figure 9: GeoPandas Visualization: Tata Motors Service Centres Overlaid on Kolkata Map

Interpretation: Figure 9 uses GeoPandas and Plotly for geographic overlay. The visualization highlights clusters in Rajarhat, Beliaghata, and Ballygunge, supporting potential optimization for demand-based service expansion.

3.4 Data Analysis of Petrol Pumps, Workshops and Dhabas

3.4.1 Infrastructure Correlation



Dhabas and Workshops Show Strong Association

From the correlation heatmap, the correlation values are:

Infrastructure Pair Correlation

Petrol Pumps – Dhabas 0.47

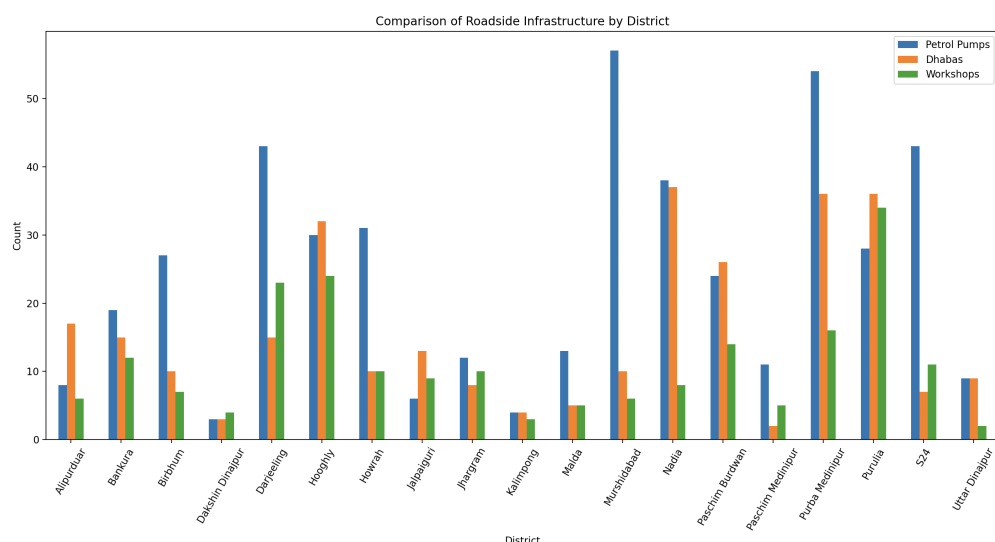
Petrol Pumps – Workshops 0.41

Dhabas – Workshops 0.68

Dhabas and workshops tend to appear together more often than petrol pumps and workshops.

The correlation analysis reveals that dhabas and workshops have the strongest positive relationship (correlation 0.68). This suggests that food facilities and vehicle repair services tend to develop together along transportation routes, likely due to shared demand from long-distance travelers and commercial transport vehicles.

3.5 Service Distribution



1. Petrol Pumps are the Dominant Infrastructure: From the bar chart, petrol pumps are consistently higher in number than dhabas and workshops across almost all districts

District	Petrol Pumps	Dhabas	Workshops
Murshidabad	57	10	6
Purba Medinipur	54	36	16
Darjeeling	43	15	23

Table 2: Comparison of roadside infrastructure in selected districts

2. Across most districts:

Petrol Pumps $\gg$ Dhabas $\gg$ Workshops

- (a) Murshidabad: 57 petrol pumps vs only 6 workshops
- (b) Paschim Burdwan: 24 petrol pumps vs 14 workshops
- (c) Birbhum: 27 petrol pumps vs 7 workshops

Vehicle servicing facilities are significantly fewer than fueling stations.

3. . District-Level Service Hubs

Some districts exhibit high numbers of all three infrastructure types, suggesting they function as regional service hubs.

Major service hubs visible in the graph

High infrastructure districts include:

- (a) Purba Medinipur
- (b) Murshidabad
- (c) Darjeeling
- (d) Nadia
- (e) Hooghly
- (f) Purulia

These districts likely correspond to major transportation corridors or high traffic regions.

4. Underserved Districts Some districts show very limited service infrastructure.

District	Petrol Pumps	Dhabas	Workshops
Kalimpong	4	4	3
Dakshin Dinajpur	3	3	4
Jalpaiguri	6	13	9

Table 3: Roadside infrastructure in low service density districts

These districts may face limited vehicle support services, especially for long-distance travelers.

3.6 References and Data Sources

- Market stats: <https://auto.hindustantimes.com>
- Market shares: <https://factodata.com/car-market-share-in-india-2025>
- Tata sales stats: <https://news.jobaaaj.com/tata-motors-sales-october-2025>
- Dealer and service center details: <https://cardekho.com/15-tata-car-showrooms-in-kolkata>, <https://carwale.com/tata-showrooms-in-kolkata>
- Charging infrastructure Kolkata: <https://energytransitionkolkata.teri.res.in/>, <https://e-amrit.niti.gov.in/charging-map>
- CSV datasets: <https://scrapehero.com/report/Tata-Motors-India>, <https://kaggle.com/datasets/pranjalsingh25/ev-charging-stations-in-india>

3.7 Service Center Expansion

Despite robust sales, service infrastructure has not scaled proportionally (details and trends in market report [?]). Company announcements suggest select players are expanding networks, but overall coverage remains a fraction of the need.

This disparity is the core of the problem statement.

4 Two-Stage Mathematical Formulation

To balance profitability with customer service efficiency, a two-stage (lexicographic) mixed-integer linear programming (MILP) model is proposed. The first stage maximizes the overall profit from service operations, while the second stage minimizes the total service time subject to maintaining the profit obtained in Stage 1. This approach ensures that operational efficiency is optimized without compromising profitability.

4.1 Stage 1: Profit Maximization Model

Sets and Indices

- I : Set of demand zones (indexed by i)
- J : Set of existing service centres (indexed by j)
- K : Set of candidate new service centres (indexed by k)
- C : Set of EV charging stations (indexed by c)

Parameters

- D_i : Service demand at demand zone i
- $d_{i\ell}$: Distance (km) between zone i and centre $\ell \in (J \cup K)$
- C_k^{new} : Service capacity of candidate centre k
- C_j^{old} : Service capacity of existing centre j
- S_j^{base} : Baseline service requirement of existing centre j
- α : Minimum service fraction for incumbents ($0 \leq \alpha \leq 1$)
- $\bar{p}$: Average billing per service (INR/service)
- v : Variable cost per service (INR/service)
- $P = \bar{p} - v$: Net profit per service (INR/service)
- F_k : Fixed annual cost to open new centre k
- λ : Distance penalty (INR per km per service)
- B : Budget for new centre establishment (INR)
- M : Maximum number of new centres that can be opened
- R_{max} : Maximum allowable service radius (km)

Decision Variables

$$x_k = \begin{cases} 1 & \text{if new service centre } k \text{ is opened,} \\ 0 & \text{otherwise} \end{cases}$$

y_{ik} : Services allocated from zone i to new centre k

z_{ij} : Services allocated from zone i to existing centre j

Objective Function (Stage 1)

$$\begin{aligned} \max Z_1 = & \sum_{i \in I} \sum_{k \in K} P y_{ik} + \sum_{i \in I} \sum_{j \in J} P z_{ij} - \sum_{k \in K} F_k x_k \\ & - \lambda \sum_{i \in I} \sum_{\ell \in (J \cup K)} d_{i\ell} a_{i\ell} \end{aligned} \quad (1)$$

where

$$a_{i\ell} = \begin{cases} y_{ik} & \text{if } \ell = k \in K \\ z_{ij} & \text{if } \ell = j \in J \end{cases}$$

Constraints

$$\sum_{k \in K} y_{ik} + \sum_{j \in J} z_{ij} = D_i \quad \forall i \in I \quad (2)$$

$$\sum_{i \in I} y_{ik} \leq C_k^{\text{new}} x_k \quad \forall k \in K \quad (3)$$

$$\sum_{i \in I} z_{ij} \leq C_j^{\text{old}} \quad \forall j \in J \quad (4)$$

$$\sum_{i \in I} z_{ij} \geq \alpha S_j^{\text{base}} \quad \forall j \in J \quad (5)$$

$$\sum_{k \in K} F_k x_k \leq B \quad (6)$$

$$\sum_{k \in K} x_k \leq M \quad (7)$$

$$a_{i\ell} = 0 \quad \forall (i, \ell) : d_{i\ell} > R_{\max} \quad (8)$$

$$x_k \in \{0, 1\}, \quad y_{ik}, z_{ij} \geq 0 \quad (9)$$

The optimal value obtained from Stage 1 is denoted as Z_1^* and represents the maximum achievable profit.

4.2 Stage 2: Service Time Minimization Model

Once the maximum profit Z_1^* is obtained, the second stage minimizes total service time while ensuring that the profit does not drop below a specified tolerance δ (e.g., 0.1% below the optimal value).

Additional Parameter

$t_{i\ell}$: Average service time (hours per service) from zone i to centre ℓ

δ : Acceptable profit tolerance (e.g., 0.001 for 0.1%)

Objective Function (Stage 2)

$$\min Z_2 = \sum_{i \in I} \sum_{\ell \in (J \cup K)} t_{i\ell} a_{i\ell} \quad (10)$$

Constraints

All constraints from Stage 1 (2)–(8) are retained, with an additional constraint to preserve near-optimal profit:

$$Z_1 \geq (1 - \delta) Z_1^* \quad (11)$$

Remarks

- Stage 1 ensures economic feasibility and determines which centres to open.
- Stage 2 enhances operational efficiency by minimizing total service time or response distance.
- Both stages are linear, ensuring computational tractability.

Solution Approach

The two-stage optimization is implemented in Python using the **Gurobi** optimizer. The procedure is as follows:

1. **Stage 1:** Solve the profit maximization model (1)–(8) to obtain Z_1^* and the corresponding binary variables x_k^* indicating which centres are opened.
2. **Stage 2:** Fix $x_k = x_k^*$, add constraint (11), and minimize total service time (10).
3. The resulting allocations $\{y_{ik}, z_{ij}\}$ represent the optimal service distribution ensuring minimal average service time without sacrificing profitability.

4.3 Computation of Inter-Location Distances

To accurately represent the spatial relationship between demand zones and service centers, the distance between each pair of locations (i, s) is computed using the **Haversine formula**, which calculates the great-circle distance between two points on the Earth's surface based on their geographic coordinates.

Haversine Distance Formula:

$$a = \sin^2\left(\frac{\phi_s - \phi_i}{2}\right) + \cos(\phi_i) \cos(\phi_s) \sin^2\left(\frac{\lambda_s - \lambda_i}{2}\right) \quad (12)$$

$$d_{is} = 2R \arcsin(\sqrt{a}) \quad (13)$$

where:

- ϕ_i, ϕ_s are the latitudes (in radians) of locations i and s .
- λ_i, λ_s are the corresponding longitudes (in radians).
- $R = 6371$ km is the average radius of the Earth.

This formula ensures that the computed distances d_{is} reflect realistic travel distances between locations across Kolkata, accounting for the curvature of the Earth.

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5 Conclusion

The study presents a comprehensive framework for optimizing automotive service network coverage in urban India, focusing specifically on the case of Tata Motors service centers in Kolkata. By combining real-world location data, advanced mathematical programming, and spatial analytics, this report demonstrates a scalable methodology to bridge the gap between rapid vehicle growth and lagging service infrastructure.

Key findings show that existing service centers do not sufficiently match the pace or spatial demands of increasing vehicle ownership, but strategic expansion can significantly improve both profitability and operational efficiency. The mixed-integer linear programming approach successfully balances economic objectives—such as maximizing net profit and minimizing response time—with practical constraints like capacity, service radius, and budget.

The results, including marginal analyses and integrated allocation maps, provide actionable insights for both practitioners and policymakers. The approach ensures customer needs are met more equitably, and resources are allocated in a manner that maximizes system-wide benefit.

Overall, this work illustrates the critical importance of data-driven optimization in transport service management and lays the groundwork for continued research in service infrastructure planning, especially as new technologies and mobility patterns emerge.

6 Future Aspects

The results and methodology of this report highlight several opportunities for further research, industry innovation, and policy formulation in automotive service network optimization:

- **Integration of Real-Time Data:** Future approaches should incorporate live data streams (e.g., GPS, IoT sensors, online customer feedback) to dynamically optimize service center operations, monitor traffic conditions, and reduce service response times.
- **Expansion for Electric Vehicles (EVs):** As the share of EVs in urban fleets increases, coupling service center placement with charging infrastructure planning will become essential. Models may include battery health analytics, charger utilization rates, and forecasted demand for EV-specific maintenance.
- **Regional Customization and Scalability:** The modeling framework can be extended to other cities and regions with different demographic, economic, or geographic characteristics. Scalability to state or national levels—and adaptation for both dense urban and remote rural areas—will improve accessibility and equity.
- **Multi-Objective and Robust Optimization:** Future research should address additional objectives such as minimizing carbon footprint, maximizing customer satisfaction, or integrating risk factors (e.g., disaster resilience, supply chain disruptions) into location and resource allocation strategies.
- **Machine Learning for Predictive Demand:** Advanced machine learning models can be employed to predict demand peaks, customer churn, and service failures, enabling proactive interventions and improved resource allocation.
- **Automated Decision Support Systems:** Combining mathematical optimization with interactive dashboards and visualization tools will empower managers and local planners to simulate scenarios and make evidence-based decisions in real time.
- **Policy and Regulatory Innovation:** Collaboration between industry, government, and academia will help define new standards for service quality, regulatory compliance, and incentives for expanding coverage in underserved areas.

Summary: Continued innovation and research in automotive service network design will play a critical role in supporting India’s vehicle growth, ensuring safety, satisfaction, and sustainability for millions of road users.

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